Research on a BIM Model Quality Compliance Checking Method Based on a Knowledge Graph

Simiao Zhu[1]; Jianliang Zhou[2]; Lei Cheng[3]; Xuehai Fu[4]; Yan Wang[5]; and Kexin Dai[6]

Abstract: With the development of information technology, building information modeling (BIM) is becoming widely used in the construction industry. It is very important to ensure the quality and compliance of BIM models. To solve the problem of time-consuming and laborious traditional checking methods and the limitation that existing software hard coding cannot adapt to rule changes, this paper proposes a BIM model quality compliance checking method based on knowledge graph that can quickly check BIM models through automated checking and ensure the modeling quality of BIM models. Firstly, knowledge extraction for BIM modeling standards and delivery standards is carried out based on knowledge graph technology to structurally represent text. Then, Revit secondary development technology and the C# programming language are adopted to develop a BIM model quality compliance checking system based on the knowledge graph analysis results and the structured query language (SQL) built-in rule knowledge base. Based on the knowledge base, the BIM model is automatically checked for quality compliance, and the naming, material, parameters, and rationality of each professional component are checked. Next, it is checked whether it conforms to the standard. Finally, the model modification plan is provided by checking the problem components and noncompliance details. The case application results show that the BIM model quality compliance checking system developed based on knowledge graph technology can identify defects or errors that do not meet the standards in the model, verify the feasibility and practicability of the system, and provide a new method for BIM designers to modify and improve the model. **DOI: 10.1061/JCCEE5.CPENG-5950.** © *2024 American Society of Civil Engineers.*

Author keywords: Knowledge graph; Building information model; Automated checking; Rule base; C#; Revit secondary development.

Introduction

With the rapid development and widespread application of building information modeling (BIM) technology, the construction industry is entering an era of digital transformation. BIM technology has received high attention from the state in China, and a series of BIM policies have been introduced one after another to promote the development of BIM technology in China. The market scale for building information models in China is increasing year by year, and enterprises' investment in BIM applications has shown a growing trend. More and more construction projects adopt a BIM model for design delivery.

In the application process of BIM technology, the compliance of the model is crucial. The quality of BIM models is significant for the accuracy of the entire project (Wangara 2018). BIM model quality compliance checking is a technical method to evaluate the quality compliance of building information models based on BIM modeling standards (Choi and Lee 2022). However, due to the different usage habits of different designers, the property settings of various components in the BIM model are also different. In the process of creating models, many modeling software programs allow the use of common component libraries and custom families, which can lead to modeling errors and omissions (Koo and Shin 2018). Moreover, existing BIM models suffer from problems such as low standardization and uneven quality levels, which lead to poor information exchange among construction parties due to nonstandard components. Therefore, there is an increasingly urgent need for compliance checking of BIM models.

Quality checking of BIM models involves verifying that the model has been correctly designed and defined, and that the model components are complete or otherwise required (Wangara 2018). Traditional model checking relies heavily on human experience, which is time-consuming and laborious. In fact, since the concept of automated compliance checking (ACC) was proposed, many studies have used hardcoding to turn standards and specifications into computer-executable rules to realize automated compliance checking, but this approach is less flexible and less interoperable, and it is difficult to adapt to the ever-changing rule requirements (Solihin and Eastman 2016).

In contrast, a knowledge graph is a kind of network containing rich semantic relations that can integrate data from different

[1]Research Assistant, School of Mechanics and Civil Engineering, China Univ. of Mining and Technology, Jiangsu, Xuzhou 221116, China. Email: zhusimiao@cumt.edu.cn

[2]Professor, School of Mechanics and Civil Engineering, China Univ. of Mining and Technology, Jiangsu, Xuzhou 221116, China (corresponding author). ORCID: https://orcid.org/0000-0002-4196-3247. Email: zhoujianliang@cumt.edu.cn

[3]Research Assistant, School of Mechanics and Civil Engineering, China Univ. of Mining and Technology, Jiangsu, Xuzhou 221116, China. Email: TS23030135P31@cumt.edu.cn

[4]Research Assistant, School of Mechanics and Civil Engineering, China Univ. of Mining and Technology, Jiangsu, Xuzhou 221116, China. Email: fu_xuehai@cumt.edu.cn

[5]Ph.D. Student, School of Mechanics and Civil Engineering, China Univ. of Mining and Technology, Jiangsu, Xuzhou 221116, China. Email: 11788@xzit.edu.cn

[6]Research Assistant, School of Mechanics and Civil Engineering, China Univ. of Mining and Technology, Jiangsu, Xuzhou 221116, China. Email: TS21030142P31@cumt.edu.cn

Note. This manuscript was submitted on January 17, 2024; approved on July 9, 2024; published online on October 17, 2024. Discussion period open until March 17, 2025; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Computing in Civil Engineering***, © ASCE, ISSN 0887-3801.

sources, correlate all kinds of information, allow the reuse of information (Pauwels et al. 2010), and fully express the constraint relations between entities, providing a new way of thinking for compliance checking. Due to the large volume of construction projects, long construction cycle, complex unstructured data, and different BIM delivery standards required by different projects, it is necessary to apply knowledge graphs in the process of BIM model checking for effective expression of complex relationships in the rules of the provisions to achieve a comprehensive modeling quality of BIM model analysis (Deng et al. 2022).

To solve the problem of time-consuming and laborious traditional checking methods and the limitations of hard coding that cannot adapt to the change of rules, a method of BIM model quality compliance checking based on knowledge graphs is proposed in this study, and a soft coding form is provided to realize rule compilation. The BIM model automatic check plug-in generated by the secondary development of Revit based on C# has strong practical value. The contents of this study include the following:

- The knowledge graph is established by extracting the knowledge of BIM modeling standards and delivery standards.
- A dynamic rule base and BIM model quality compliance checking system are established based on the algorithm.
- The feasibility and practicability of the developed system are verified by a case study.

This study can help ensure the quality and accuracy of building information models, improve work efficiency, and reduce errors and duplication of work. With automated checking, potential design problems and conflicts can be quickly identified, saving time and money.

Literature Review

Quality Compliance Checking of BIM Models

In the process of BIM application, modelers need to follow the modeling standards to create building information models (Koo and Shin 2018). Contents to be checked include component naming, parameter compliance, correct spatial relationships of model components, completeness of parameters, and proper delivery formats. A BIM model is a digital modeling tool used in construction projects, integrating various types of information such as geometric shapes, spatial relationships, materials, structures, and so on. Conducting quality compliance checks on the BIM model helps confirm whether the model meets the relevant specifications, standards, and requirements. These checks are used to identify errors, omissions, conflicts, and inconsistencies in the models, thereby improving the reliability, accuracy, and coordination of the project.

The compliance check of the BIM model is actually a rule-based check (Luo and Gong 2015). Eastman et al. (2009) carried out the research on rule-based automated checks in architectural design. They proposed four stages for rule-based checks:

1. Rule interpretation: translating standard terms into computer-readable forms.
2. Model preparation: building models containing all kinds of information (types, attributes, and so on) required for rule checking.
3. Rule execution: apply rules to check the model.
4. Check report: report the check results in the form of an example graphic report or reference to the original rules.

Software development for automated specification checking first started in 1966, when Fenves (1966) discovered the decision table and worked on how to check the specification through computer programming. Singapore developed an automated code checking system, CORENET e-PlanCheck, based on the automated checking process for BIM (Liebich et al. 2004; Solihin et al. 2017). At the beginning of the 21st century, the International Code Council (ICC), as a developer and promoter of building codes, launched the SMARTcodes project, aiming to develop code interpretable software for evaluating BIM models [D. Conover, "Method and apparatus for automatically determining compliance with building regulations,"U.S. Patent Application 12/232,841 (2009)]. It is undeniable that existing software is successful in checking a specific provision. However, due to the specificity of the hard-coded rules, these software or systems can only work in specific domains or proprietary models, and their scalability and transparency of the rules need to be improved, hence the so-called black box approach (Nawari 2019).

The process of quality compliance checking of BIM models is based on BIM data and enforcing rules to enable automated checking and evaluation of the models. Solihin et al. (2017, 2020) argued that data availability and accessibility are the keys to automated rule checking, based on which they proposed a simplified BIM-based rule (BIMRL) approach for building data to achieve the optimal application of a BIM-based rule checking system. (Choi et al. 2014) developed an automated evacuation rule checking system that automatically checks and evaluates the BIM data to determine whether the model complies with the regulations. Xie et al. (2022) developed a Knowledge Base Management System (KBMS) to extract the model information data. The developed Revit plug-in has the functions of model completeness, view positioning checking, and so on. Automatic rule checking in the mechanical, electrical and plumbing (MEP) system was realized.

Currently, many researchers have begun exploring methods for quality compliance checking of BIM models, and there are many worthy research results in the fields of water supply systems, temporary facilities, modular buildings, deep foundation pits, and others. Martins and Monteiro (2013) developed a LicA application, which includes hydraulic analysis, code checking and view output, and so on. Based on BIM, it can perform automatic code checks on domestic water supply system in Portugal. Choi and Lee (2022) developed a quality checking system for temporary engineering based on BIM technology, with the main modules being BIM modeling and quantity detection. By automatically executing quality checking tasks, the quality checking process of temporary works can be improved, and the efficiency and accuracy of quality checking can be improved.

Sun and Kim (2022) proposed an automatic code compliance checking method to verify the quality of the modular BIM model. The automatic checking system established can serve as a platform for connecting users and manufacturers. Luo and Gong (2015) proposed an automated code compliance checking method based on the BIM model that can be used to carry out efficient and accurate code compliance checking in the field of deep foundation pits specification compliance checking. Koo and Shin (2018) formulated a support vector machine (SVM) framework using multiple BIM models for identifying BIM elements in BIM models and misclassifying of industry foundation class (IFC) classes.

Application of Knowledge Graph in Automatic Checking

The concept of a knowledge graph was formally put forward by Google in 2012 to support semantic searching. By utilizing knowledge graph technology, industry knowledge can be continuously classified and summarized into different knowledge ontologies in the field of construction engineering. Common techniques for automatic code compliance checks include the following

(Guo et al. 2021; Ismail et al. 2017): (1) using existing software or plug-ins for code compliance checks; (2) establishing a knowledge base and rule engine based on object-oriented methods; (3) interpreting rules based on logical methods such as predicate logic, concept maps, and deontology; and (4) creating an ontology method framework based on semantic web technologies.

Existing BIM check tools mostly rely on hard-coded rules (Bloch and Sacks 2020), which have the main drawbacks of limited scalability and high maintenance and modification costs, making them unable to adapt to the continuous revision of building codes. Users of such software can only customize rules based on the provided rule templates and project requirements, without the ability to access, modify, or adjust the system's rules (Fan et al. 2019). Therefore, the goal of developing an automatic checking platform capable of accommodating all building regulations remains challenging (Bloch and Sacks 2020).

Knowledge graph technology makes it possible to develop compilation rules in the direction of soft coding. Based on the knowledge graph technology, the standards of the construction industry can be updated and managed in real time, and when the standards change, the knowledge graph can be updated in time to ensure the accuracy and compliance of the checking. Quality compliance checking through the knowledge graph ensures consistency between BIM models from different teams and different phases, enabling each team to model and check according to the same requirements.

In view of the limitations caused by hard coding of normative texts, many scholars have carried out research on converting rules into soft-coding forms, including semantic web, ontology, machine learning, natural language processing, and other methods. Jouve et al. (2003) and Esteves and Joseph (2008) used semantic web and ontology technologies, respectively, to extract knowledge from government regulation texts and conduct compliance checks based on knowledge models. Solihin and Eastman (2016) used knowledge representation to formalize rules and capture rules in the form of concept maps. Zhang and El-Gohary (2016, 2017) proposed an automated compliance checking method for building codes based on natural language processing technology. Salama and El-Gohary (2016) proposed a text classification algorithm based on semantic and machine learning to realize automatic rule extraction of regulations and contract specifications. Zhou and El-Gohary (2017) proposed an information extraction algorithm based on ontology technology to realize automatic energy compliance checking in the building field. Zhang and El-Gohary (2021) and Wang and El-Gohary (2022) extracted building code relationships based on machine learning methods to support automatic compliance checking onsite.

Through the knowledge representation method of semantic modeling, dynamic rules can be generated for different building codes. Yang et al. (2023) established a new representation model based on knowledge graph, designed a semiautomatic construction design code and automatic bidding process, and verified the feasibility of the model and method through case studies and experiments. Peng and Liu (2023) based on BIM and knowledge graph technology, transformed the standard clauses into a graph database and created an automatic code compliance check system, enabling automatic checking for BIM models and the generation of checking reports. Wang et al. (2022) developed a plug-in to extract BIM model information into a graph database, and translated MEP specifications into a knowledge base based on knowledge graph technology, so as to realize model checking and rule compliance checking of the MEP system. Beach et al. (2015) developed a rule-based semantic approach and tested a regulatory compliance system framework to demonstrate its generality.

Knowledge graphs have been studied more extensively for automated checking of construction safety. Zhang et al. (2015) formalized building safety codes and established three models based on ontology, namely, building product model, building process model, and building safety model, to realize automatic safety planning based on BIM. Shen et al. (2022) combined natural language processing and ontology technology to establish a safety rule base and created a dynamic building safety rule checking framework, enabling automatic identification of safety risks during building construction. Li et al. (2022) integrated subway construction safety risk factors with building regulations using BIM and semantic web methods, developed an ontology, and performed logical reasoning to achieve automated rule checking based on SPARQL protocol and RDF query language (SPARQL). Bao et al. (2022) developed a dynamic safety check framework based on rules and topological structures. By extracting the attribute information from the BIM model, constructing a topological relationship database, and executing rule checks, automatic safety checks can be achieved.

This synthesis of existing research shows that a knowledge graph–based BIM model quality compliance checking method has important research value and application prospects. Most of the existing research is based on the normative checking of building codes, and there are fewer studies on the automated compliance checking of models for BIM modeling standards. China's BIM standards or specification system is complex; the text presents the characteristics of multisource heterogeneity. Knowledge graph technology can parameterize and structure the text information and realize the mutual mapping of spatial attribute information between BIM standards and BIM model elements through information extraction and knowledge base construction. This study introduces the emerging knowledge engineering technology of knowledge graphs and uses natural language processing (NLP) technology to structure the entities, concepts, events, and their relationships in the specification, build a rule base, and realize the knowledge expression of BIM standard clauses. Then, through the establishment of an automated compliance checking system for BIM models based on algorithms, automated checking and quality improvement of BIM models are realized.

Methodology

BIM model quality compliance checking, different from building code compliance checking, refers to checking and evaluating the quality of the building information model at the design phase and the construction phase, thereby judging whether the created model meets the requirements based on the national, industry, and enterprise BIM standards or BIM delivery standards. Conducting BIM model quality compliance checking can timely discover and correct design irrationalities and realize high-quality delivery of design results. The purpose of this study is to analyze the rules of BIM modeling standards based on knowledge graph technology, and to realize the automatic quality compliance checking of BIM models by developing program execution rules. The research framework of this paper is shown in Fig. 1.

The four execution rules are as follows:

1. Rule interpretation: Based on knowledge graph technology, knowledge extraction of BIM modeling standards is carried out, knowledge representation is carried out, and rule knowledge base is established by structured query language (SQL) language.
2. Model preparation: The BIM model is created with Revit version 2020, and information is extracted from the model based

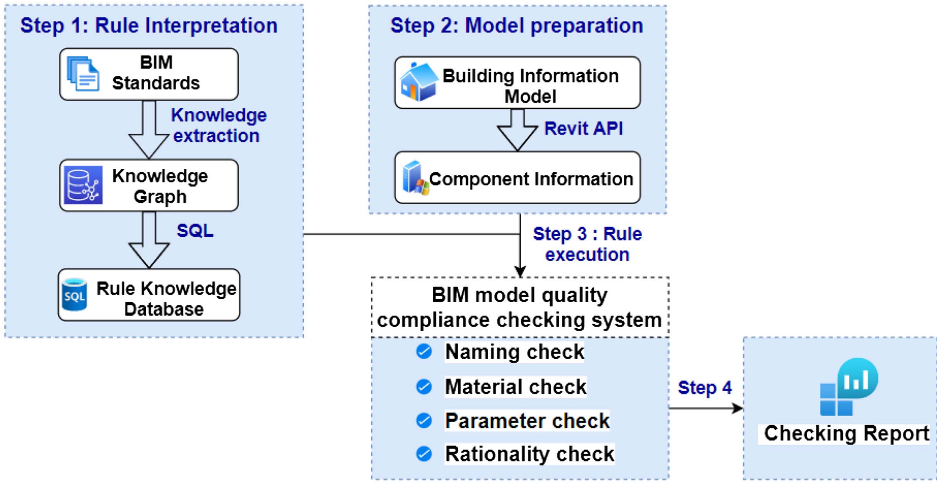


Fig. 1. Research framework of BIM model quality compliance checking based on knowledge graph.

on the Revit application programming interface (API) to match the rule knowledge base.

3. Rule execution: The C# language is used to develop the Revit plug-in, and a BIM model quality checking system is established. The functions of the checking system include naming check, material check, parameter check, and rationality check. The rule knowledge base and component information are matched in the system so as to judge whether the model modeling conforms to the rules.
4. Checking report: The checking report is presented in a visual form to judge the compliance of each component of the model in detail, including the name, ID, and noncompliance details of the component. The user clicks on the report window view of the faulty component to automatically locate the faulty component for modification.

Establishment Process of Knowledge Graph for Compliance Check

A knowledge graph is a graphical model for representing and organizing knowledge. It is a semantic network composed of entities and relationships between entities (Deng et al. 2022). As a structured representation method, the knowledge graph aims to capture the associations between entities and help computers understand and infer complex semantic relationships. In engineering, knowledge graphs can be used to represent various knowledge elements such as engineering components, materials, specifications, standards, as well as their relationships and properties. The goal of this research is to standardize and semantically express the text in engineering specifications, realize the digitalization of engineering specification data, and assist users in conducting quality compliance checks on BIM models.

Data Sources

The quality compliance checking of the BIM model is based on the BIM modeling standard. To meet the needs of engineering construction, local and national BIM policies and standards have been introduced successively. The existing BIM standards for building information models have systematically elaborated on the naming, color, and classification of building components, and so on. Common BIM modeling standards include existing Chinese national BIM standards: [GB/T51212-2016 (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2016); GB/T51235-2017 (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2017a); GB/T51269-2017 (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2017b); GB/T51301-2018 (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2018); GB/T51447-2021 (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2021)]. In addition, China Architectural Design Institute (2016) provides basic rules for the naming of files, systems, components, and so on.

These standards have comprehensively constrained the design and delivery requirements of BIM models and are used to standardize the attributes of BIM model components to improve the readability, maintainability, and accuracy of data exchange. The standard system in China contains several levels. The enterprise delivery standard for the BIM model is based on the Chinese national standard.

Therefore, the data of the knowledge graph created in this study is based on the aforementioned national BIM standards, combined with the BIM delivery standards of the enterprise. Based on the characteristics of the syntactic structure, the aforementioned specifications are translated into a language that can be recognized by the computer, and knowledge extraction of the text is carried out to build the knowledge graph. The translated rules are compiled into the system by means of algorithms.

Knowledge Extraction

NLP technology can perform word segmentation and semantic analysis of normative texts, which is convenient for the coding of subsequent normative provisions (Chowdhary and Chowdhary 2020). Knowledge extraction involves entity extraction, relation extraction, and attribute extraction. Through NLP techniques, knowledge is extracted from the collected data, and the relationships among entities, attributes, and entities are extracted to form an ontological knowledge expression. Types of data can be classified as semistructured data, structured data, or unstructured data according to their structure. Natural language processing includes two mechanisms: information retrieval (IR) and information extraction (IE) (Chowdhary and Chowdhary 2020), in which IE aims to extract structured data from unstructured and semistructured text. In this research, named entity recognition (NER) technology is used to identify key entities in the text. NER technology is capable

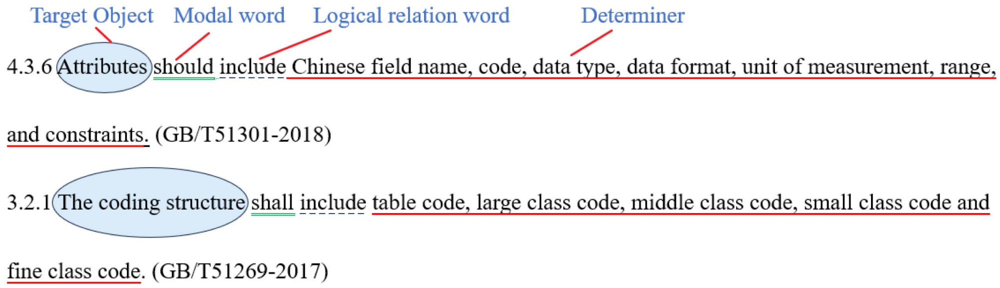


Fig. 2. Article structuring based on natural language processing.

of identifying and classifying information elements from the text, referred to as named entities (NE) (Yan et al. 2018).

Natural Language Parsing

In this study, the specifications are disassembled according to natural language processing mechanisms (including word segmentation, part-of-speech tagging, syntactic analysis, target keywords, and constraint sequence extension) (Wang and El-Gohary 2022). The structured processing of normative provisions is to classify the entities and constraint relations in text and translate them into parameterized logical rules that can be processed by computers. The normative provisions generally have the structural characteristics of target object, modal word, logical relation word, attribute word, determiner word, and so on. The basic units of specification checking are the target object, attribute word, and qualifier. Modal words are used to express the degree of rigor required by a norm (appropriate, strictly prohibited, and so on). Logical relation words are divided into judgment relation words (yes, satisfy, conform, and so on), position relation words (upper, middle, lower, and so on), comparison relation words (less than, greater than, not less than, and so on), distance relation words (distance, to, and so on), and inclusion relation words (setting, arrangement, and so on).

Taking Article 4.3.6 of GB/T51269-2017 and Article 3.2.1 of GB/T51301-2018 as examples, the target object, modal words, logical relation words, and qualifiers can be determined (Fig. 2). By analogy, in a similar way, the structural rules in the normative provisions are found, the entity and attribute relations are extracted, and the text of the standard is translated to prepare for the construction of the knowledge graph.

Named Entity and Attribute Recognition

In the context of BIM, knowledge maps are constructed by identifying and relating building elements, such as walls, floors, and so on, to create nodes. These nodes can not only represent specific building components but also further enrich their data content by defining properties such as materials, dimensions, and others. In addition, relational extraction technology is used to identify and extract semantic relationships between entities in text, which can be relationships between entities or relationships between entities and attributes. From these relationships, we can construct triples (subject-predict-objects), which are the basic units of the knowledge graph.

To combine the component information of the BIM model, we have carried out in-depth sorting and analysis of BIM standard documents. Through this process, we can tease out key knowledge elements such as entity elements, attribute associations, and geometric data in the model. Further, we systematically extract and classify BIM models (Fig. 3), including but not limited to the following categories:

- Professional classification: classify components according to different professional areas of the construction industry.

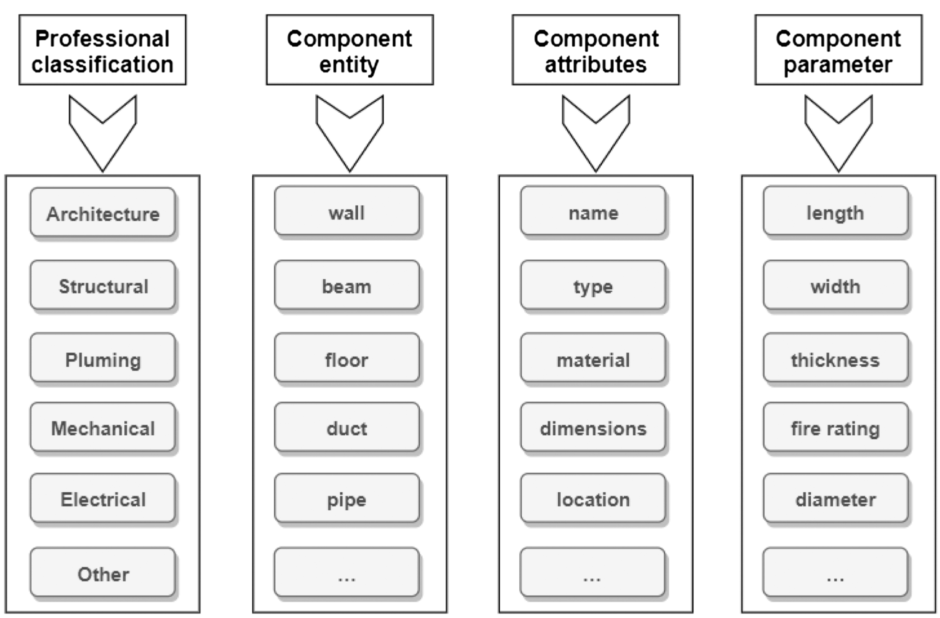


Fig. 3. Knowledge extraction and classification.

Table 1. Identification of entities and attributes of BIM standard provisions

Profession	Entity	Attribute	Parameter
Architectural	Wall	Bottom elevation	Top of structural plate
		Top elevation	Bottom of structural plate

- Component entity: identify and classify specific components in the BIM model.
- Component attributes: define the attributes of components, such as material, size specifications, and so on.
- Component parameters: list the detailed parameters of the component, which are critical to the function and performance of the component.

Taking the clause "the bottom of the building wall is attached to the top of the structural board, and the top of the wall is attached to the bottom of the structural board" as an example, the entity and attribute information is extracted as presented in Table 1.

Knowledge Fusion

To improve the quality of the knowledge graph, data integration and entity alignment of heterogeneous data sources are needed. Therefore, data preprocessing is required to facilitate subsequent data analysis and modeling. By merging the same entity and the same relationship, unified rules are established based on normative texts, including naming rules for components, color-matching rules for materials, parameter-setting rules, and rationality rules.

Component Naming Rules

Due to the different naming habits of model creators, the names of various components in the modeling process vary greatly. Therefore, this study refers to the *Chinese Academy of Sciences BIM Standard Series - Naming Rules* and BIM model delivery standards for enterprises to formulate the naming rules of each component to standardize the naming of various components of the model. The naming formats for some components are given in Table 2.

Material Color-Matching Rules

To improve the visualization function of the model and facilitate component management, various professions need to make systematic color-matching regulations for their respective pipelines and components. The project-based BIM model delivery standards have material requirements and red, green, blue (RGB) standards for the components of different disciplines, i.e., for the rules to be followed when modeling. This study combines national standards and industry standards to summarize the requirements for color matching of component materials, such as given in Table 3.

Parameter-Setting Rules

BIM models are parametric digital models, and the various components of the model contain numerous geometric and nongeometric information, which are stored in digital form in the model database. The geometric information of BIM model components describes the geometric features of the components, including the shape, size, and position of the components. The nongeometric information of BIM model components includes information on the structure and function of the components, such as structural form, connection method, opening mode of doors and windows, and so on. To ensure that the components in the building information model can meet the corresponding design and construction requirements, these parameters need to be defined and regulated. The parameter classification of some model components is provided in Table 4.

Rationality Rules

By analyzing the existing BIM standards, the rationality rules of the model put forward requirements for model integrity, modeling methods, component connection, and spatial relationships. Therefore, when checking the BIM model, the aforementioned rules should be analyzed based on the BIM delivery standards to test the rationality of components:

1. Model integrity: Integrity refers to whether the components contained in the model are complete. In the BIM delivery standard, the modeling content of architecture, structure, mechanical, electrical, and plumbing, and so on, will be specified, and the problem of missing painting of components is not allowed.
2. Modeling method: BIM modeling standards use different modeling methods for different components. For existing families in the Revit software, common components such as walls,

Table 2. Naming format of BIM model components (part)

Naming rules	Element	Examples
[Professional Code]-[Component Name]-[Thickness]	Wall	A-external wall-300
	Floor	S-structural slab-150
[Professional Code]-[Type Name]-[Model]	Doors and windows	A-single fire door-M1521
[Professional Code]-[Component Name and Number]-[Section Size b×h]	Beam/column	S-KL1-500×600
[Professional Code]-[Pipeline Type]-[Specification]	Pipeline	P-water supply pipe(J)-DN50
[Professional Code]-[Type Name]	Air hose	M-air supply duct
[Professional Code]-[Type Name]	Cable bridge	E-BAS
[Professional Code]-[Equipment Name]-[Installation Method]	Lighting equipment	E-distribution box-concealed installation

Table 3. Material color matching requirements for model components (part)

Elements	RGB	Elements	RGB
Off-wall	(177, 200, 234)	New air duct	(0, 255, 0)
Fireproof partition wall	(170, 45, 42)	Condensing tube	(255, 0, 255)
Beam	(0, 128, 0)	Refrigerant pipe	(128, 64, 64)
Floor	(128, 128, 128)	Cooling water supply	(0, 204, 204)
Structural column	(255, 128, 0)	Automatic alarm	(255, 127, 223)
Smoke exhaust pipe	(255, 255, 128)	Water supply pipe	(0, 153, 76)
Air supply duct	(0, 0, 255)	Pressure waste water pipe	(153, 114, 0)

Table 4. Parameters of building information model components (part)

Element/category	Parameters
Wall	Wall height, thickness, structure, fire rating, sound insulation rating, and so on
Floor	Floor thickness, load grade, structure, and so on
Beam/column	Beam/column section size, structure, load grade, and so on
Window/door	Window/door type, size, fire rating, sound insulation rating, and so on
Mechanical	Air conditioning pipe size, fan type, air volume grade, and so on
Electrical	Electrical line size, switch type, load level, and so on

beams, and so on need to use the correct family creation to create models.

3. Component connection: Check whether the component connections are correct, whether the component is connected with surrounding components, and whether it conflicts with other components.
4. Spatial scope: Check whether the floor height, entrance width, wall thickness, number of windows, air conditioning equipment capacity, and so on comply with the building requirements and specifications.

Knowledge Representation

To process the canonical text into a format that can be processed by a computer, the logical relationship between the rules is established with the extracted knowledge. A graph database is a model for representing and storing graph data, consisting of nodes and edges. Each node and edge can contain attributes to describe their characteristics (Liu et al. 2021) used for storing and querying graph data. In this study, a graph database approach is chosen for representing rule-based semantic knowledge to better represent the complex relationships and semantics of BIM modeling standards. Neo4j version 1.5.8, a graph database, is chosen for storing and representing the specification standards of BIM models due to its strong operability through a web interface, enabling operations such as writing, outputting, querying, and so on. By assigning appropriate attributes to nodes and edges, it becomes convenient for subsequent queries and analyses. Thus, a triple data structure is formed using Neo4j.

For querying and analysis, attribute graph databases are commonly used. Cypher is the declarative query language for Neo4j; it can be used to write query languages to retrieve and analyze knowledge from the attribute graph in architectural specifications (Wang et al. 2022). Neo4j can present semantic relationships between entities and can be interactively queried and analyzed. The specification mentioned in the section "Data Sources" is used for knowledge extraction, and the knowledge graph is constructed. As shown in Fig. 4, it is a knowledge graph that extracts the text from GB/T51269-2017 (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2017b), in which the relation "haspartof" represents the inclusion relation and "property" represents component attributes. For the BIM project's proprietary BIM delivery standards, further knowledge extraction and rule translation will be carried out through the system in the subsequent algorithm development process.

Development of a BIM Model Quality Compliance Checking System

BIM Model Checking Process Based on Knowledge Rules Base

In this system, the quality checking process of BIM model components is shown in Fig. 5. First, the target components of the BIM model need to be selected, and the system will automatically query the BIM standards in the rule base for quality checks. After the checking is completed, the user can view the report and manually modify the properties of components that do not meet the rules to complete the checking.

BIM Model Component Information Extraction

Extracting component information from the BIM model is one of the important steps to ensuring quality compliance. To achieve interoperability and semantic consistency of data, a data mapping approach is adopted to match BIM data with a knowledge graph. To determine whether the BIM model complies with the standards, a thorough examination of both the geometric and nongeometric information of the model components is required.

Revit is one of the most commonly used software for BIM. It is a large database itself and provides an open API. In this study, Revit 2020 is used as the loading and output tool of the BIM model, and the Visual Studio 2019 platform is used. Based on Visual Studio in the C# language for Revit secondary development, a BIM model automated checking system is designed. One of the most important algorithms for BIM model automated checking is to obtain component information and extract the parameters and attributes contained in the component. The process of object acquisition and parameter acquisition is shown in Fig. 6. The basic procedures are Create Collector → Get Family Symbol → Get Family Instance → Get Parameters → Get Value.

Rule Knowledge Base Module

Based on the related content of the knowledge graph studied as previously mentioned, the standard knowledge base of the building information model is established by an algorithm. The professional specifications of architecture, structural, mechanical, electrical, plumbing, and other professional specifications correspond to the components one by one. Access objects are created based on component ID, type, name, size, and other characteristics, as shown in Fig. 7.

To facilitate user-defined rule bases, based on previous research on BIM standard rule analysis and the establishment of the knowledge graph, the relational database SQL Server version 2019 is used to establish the rule base, the C# programming language and windows presentation foundation (WPF) framework are used to create Windows, and the interaction between the database ADO. NET assembly and the data source is realized. By clicking on the Standard panel, the rule database will be displayed. In this rule database, rules can be added, deleted, and edited. The input/output (IO) library is introduced based on the C# language, which can realize the import and export of files. Through the algorithm, the BIM standard generation rule base for different enterprises can be realized. The dynamic storage of rules in the developed rule knowledge base can realize the identification of BIM delivery standards for different projects.

Now, the Edit and Add functions will be discussed as an example (Fig. 8). An (*) field indicates that the input box for the field has to be entered; otherwise, a new rule cannot be generated. Compared with traditional hardcoding, the study's knowledge base of rules is

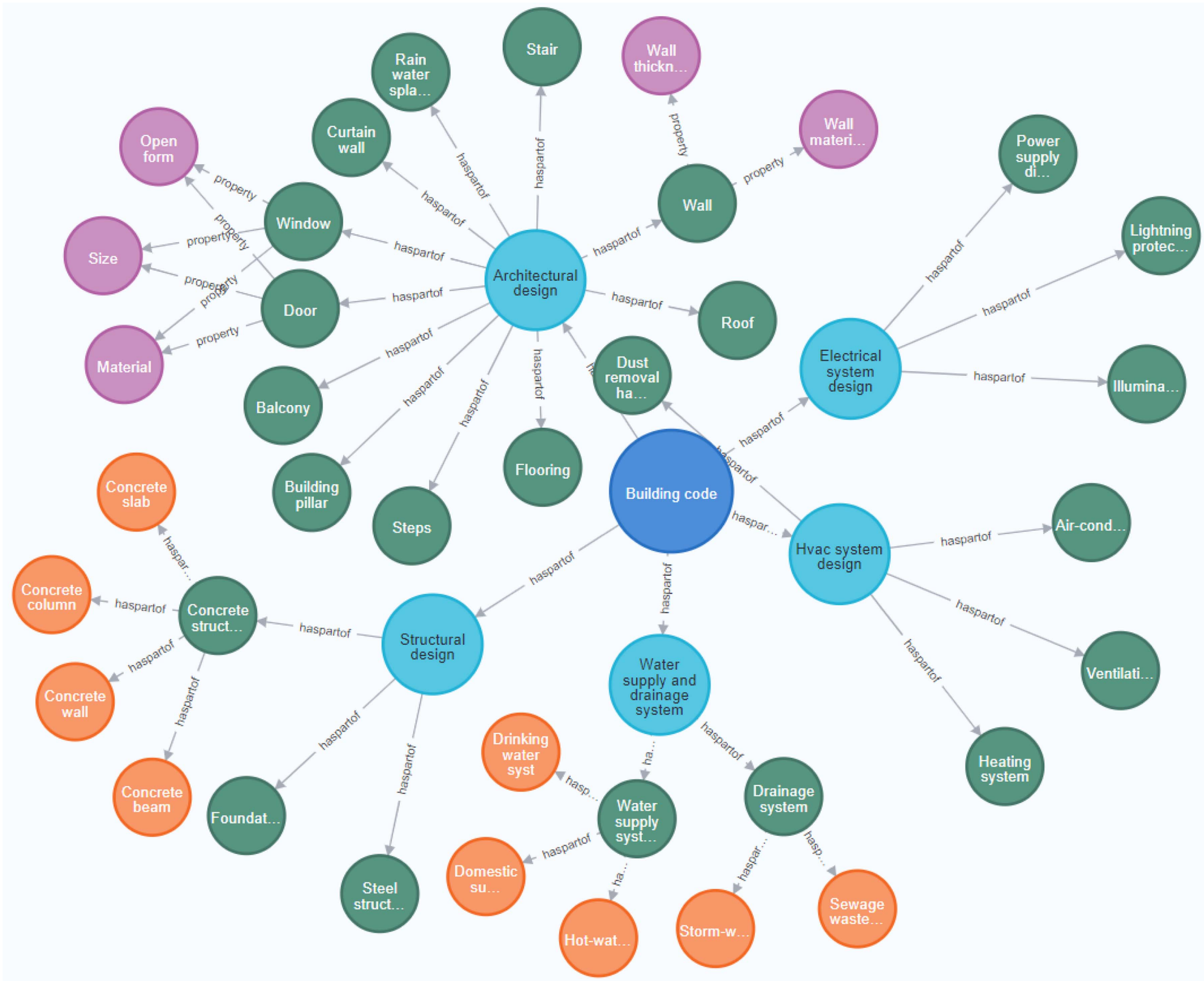


Fig. 4. Knowledge graph display based on Neo4j (partial).

stored in the database in the form of expressions of knowledge, thereby enabling the checkers to add modifications to existing rules, increasing the maintainability of the rule knowledge base. Through the dynamic rule base, different checkers can formulate model checking rules according to the needs of different BIM projects, which greatly solves the limitations of traditional hard coding. In different BIM projects, the same rules can be reused, reducing duplication of effort. Additionally, the knowledge graph–based knowledge base can be easily expanded to accommodate more rules and concepts, adapt to the size of the project, and use the SQL query language to automate compliance checks. The dynamic rule knowledge base based on knowledge graph and SQL provides a more efficient, flexible, and user-friendly solution for the compliance checking of BIM models.

Automated Checking Function Module

The automated checking function module is the core function of the BIM model quality compliance checking. Based on the rule-making process of the section “Knowledge Fusion,” the function design of this module is carried out to check whether the properties of the components in the BIM model conform to the rules. By clicking on different function panels in the system, the corresponding check function can be realized. Fig. 9 shows the functional check window of the checking system, respectively, for naming checking, material checking, parameter checking, and rationality checking.

The parts of the checking system are as follows:

1. Naming check: Check whether the naming of model components meets the standards.
2. Material check: Check whether the materials endowed by the model components meet the standards, including the naming, mapping, and color of the materials.
3. Parameter check: Check whether the parameters of the model components meet the standards, including the consistency, completeness, and accuracy of the parameters.
   - Parameter consistency: Elements of the same type in the model should have the same parameter settings to maintain consistency.
   - Parameter integrity: The parameter values of elements in the model should not be missing or incorrect. For example, the area of a room should include all the areas within the room.

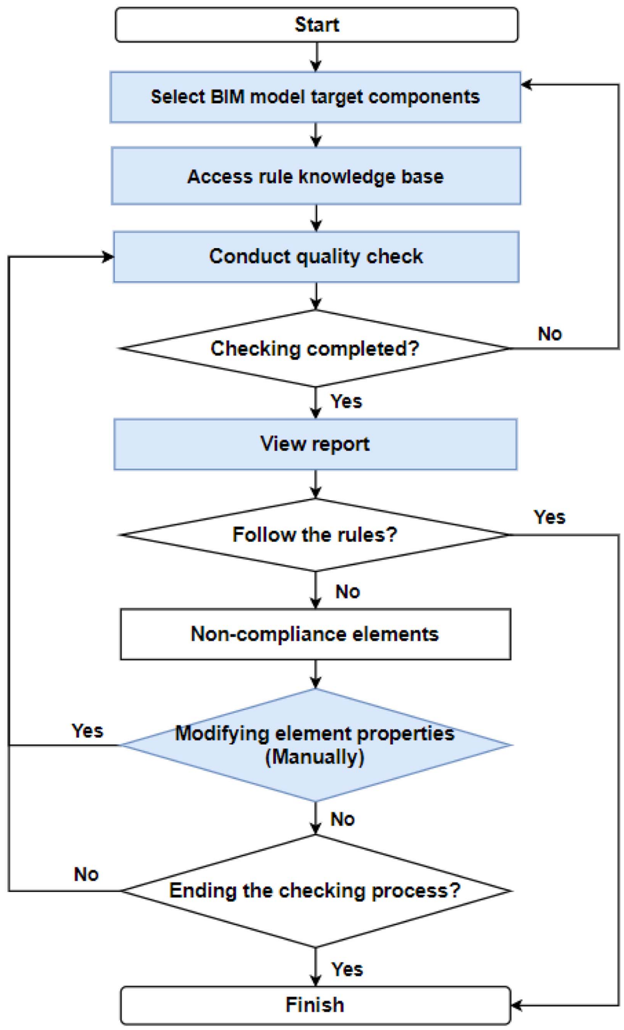


Fig. 5. BIM model checking process based on knowledge rule base.

- Parameter accuracy: The elements in the model should accurately reflect the size, shape, and position of the actual components. The size of the elements should meet the design requirements.

4. Rationality check: According to the introduction of the rationality rule in the section "Rationality Rules," this module of this system includes four parts, namely, model integrity, modeling method, component connection, and spatial scope.

Case Study

BIM Model Case Overview

The case used in this research is the Beijing–Shanghai High-Speed Railway West Station of Xuzhou Metro Line 6. The BIM model was created by the BIM engineer of the project using Revit software. The model includes the main body of the station, the attached civil construction model, the envelope structure model, the electromechanical pipeline and equipment, and the strong and weak electric pipeline and equipment models. Fig. 10 is the BIM model drawing of the station. The BIM model was checked to verify the feasibility and practicability of the developed system.

BIM Standard Rule Knowledge Base

In fact, to standardize the comprehensive BIM application of the construction drawing design stage and construction phase of each participating unit of the Rail Transit Line 6 project and to clarify the work content and deliverables of each participating unit in this stage, the BIM project manager of this project based on the existing national and local standards. In accordance with the *BIM Application Management Measures for Xuzhou Rail Transit Engineering*, the BIM delivery rules for the construction drawing design stage and construction stage of the Xuzhou Metro Line 6 project are formulated.

In this case, the naming, material, parameters, and rationality of BIM model components can be compiled by rules: (1) click Export, (2) import the BIM model delivery standard of Xuzhou Metro Line 6, (3) conduct knowledge extraction based on natural language processing technology, (4) read the naming requirements, material requirements, and parameter requirements of the BIM model from the file, and (5) add new rules to the knowledge base. The knowledge base systematically classifies the sources of BIM standards, the specific descriptions of specification texts, component entities, attributes, and checking rules, as shown in Fig. 11. The final rule knowledge base identified a total of 65 naming rules, 56 material rules, 62 parameter rules, and 43 rationality rules in the delivery standard.

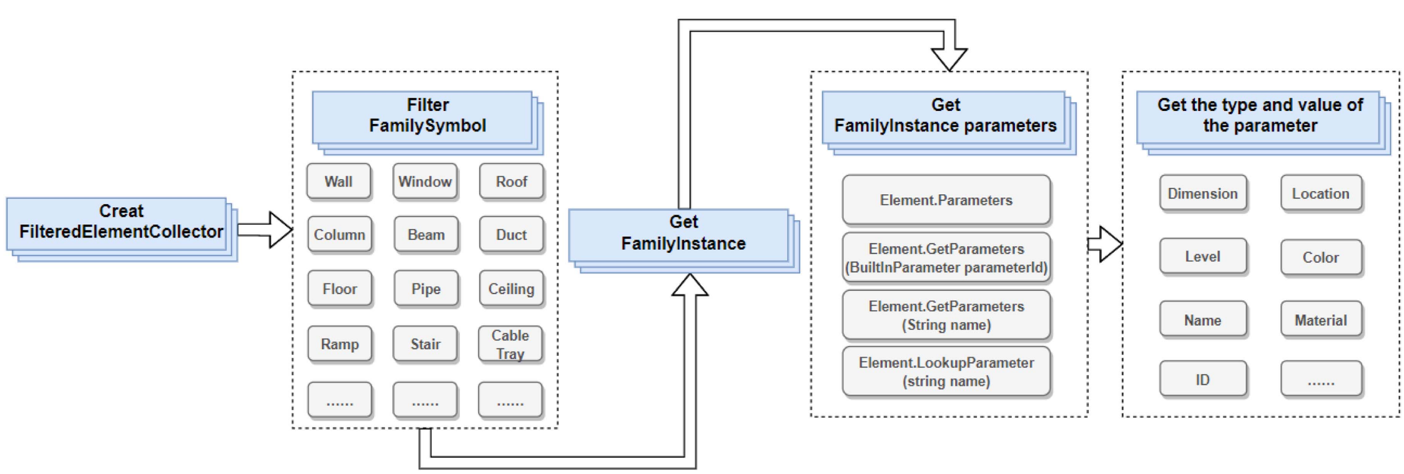


Fig. 6. BIM model component information acquisition method of Revit secondary development.

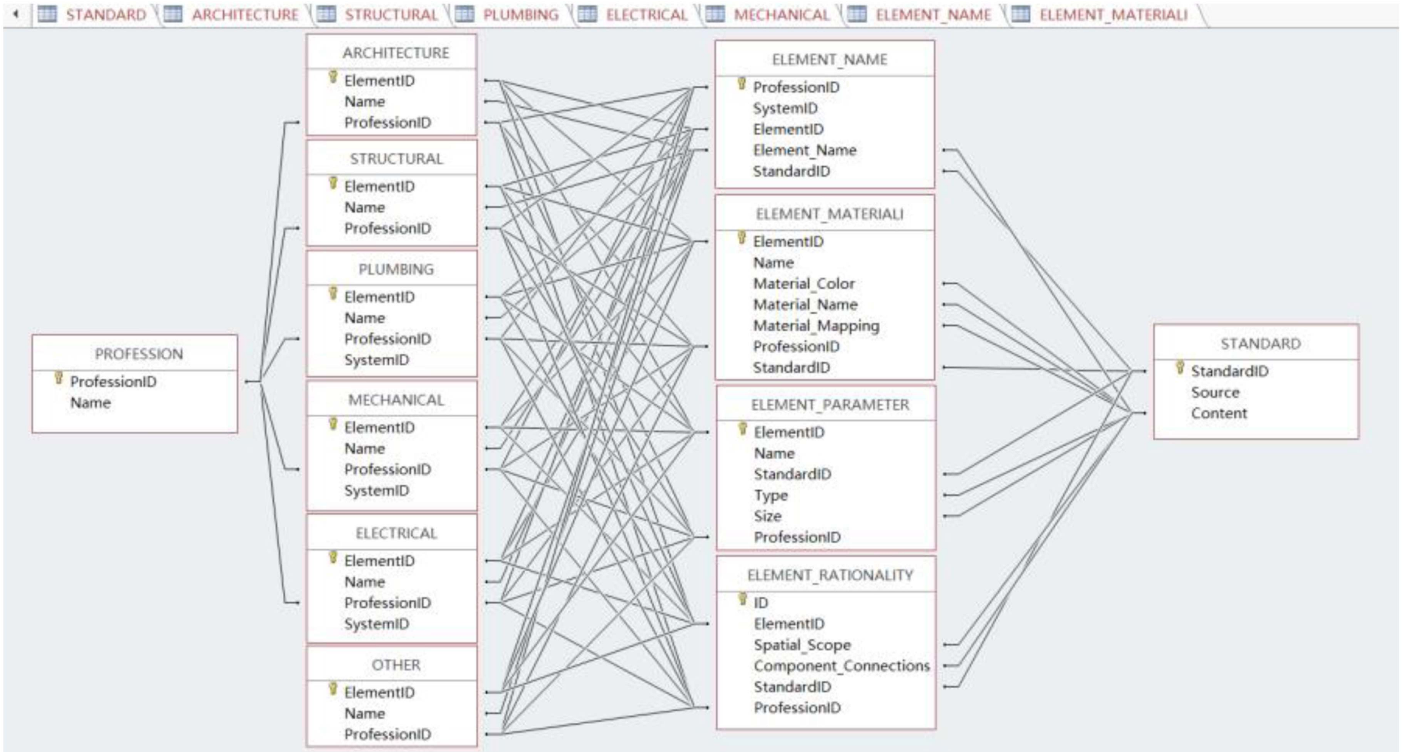


Fig. 7. Access-based rules and component information matching database.

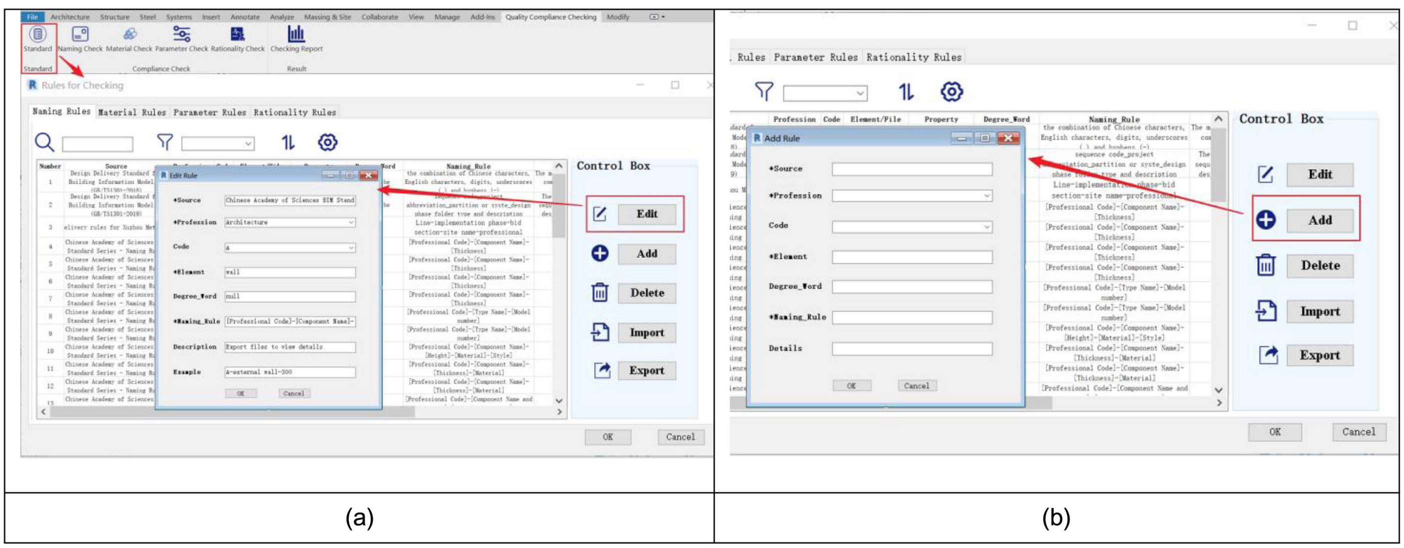


Fig. 8. Editing and add function of the rule knowledge base for the BIM model checking system: (a) editing rule function interface; and (b) add rule function interface.

BIM Model Checking Report

The checking report is the last stage of the application of the BIM model quality compliance checking system, and the BIM model checking report is designed to provide users with the ability to view model component noncompliance so that they can easily modify and improve the model. The knowledge graph–based BIM model quality checking system can functionally check all the components contained in the BIM model, and all the components of the BIM model are checked to make a judgment on component compliance. The checking results showed that there are 398 noncompliances in the model (Architecture: 79, Structural: 86, Pluming: 84, Mechanical: 53, Electrical: 47, and Other: 19).

Taking the checking results of the architectural profession as an example, the report shows the checking status of the components, including the profession to which the component belongs, the family name, the component name, the component ID, the checking content, and the details of the noncompliance. The window of the checking results and the number of problematic components in the profession are displayed in real time (Fig. 12). Therefore, users can modify the noncompliance details of problematic components to

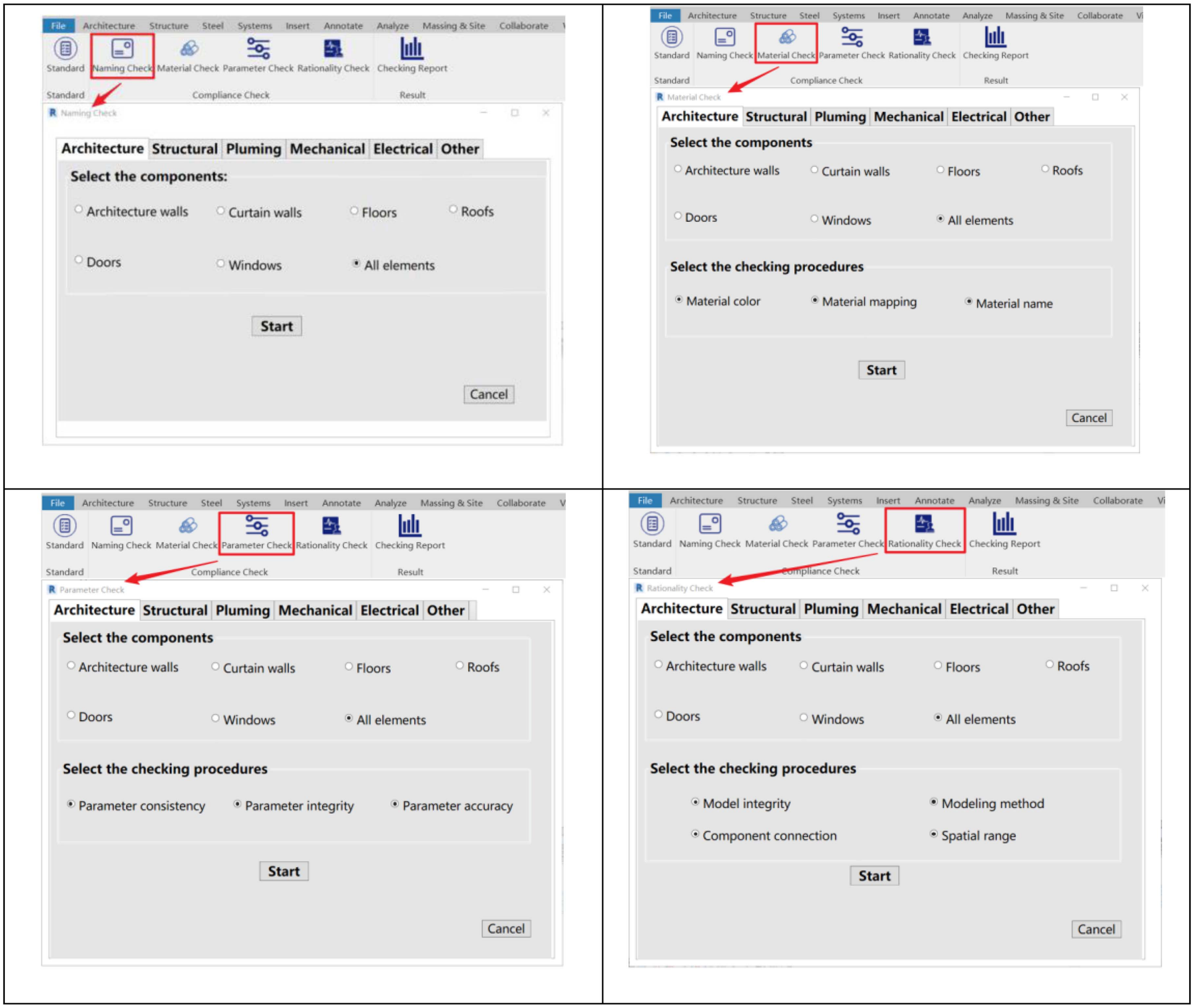


Fig. 9. Functional check windows of the BIM model quality compliance checking system.

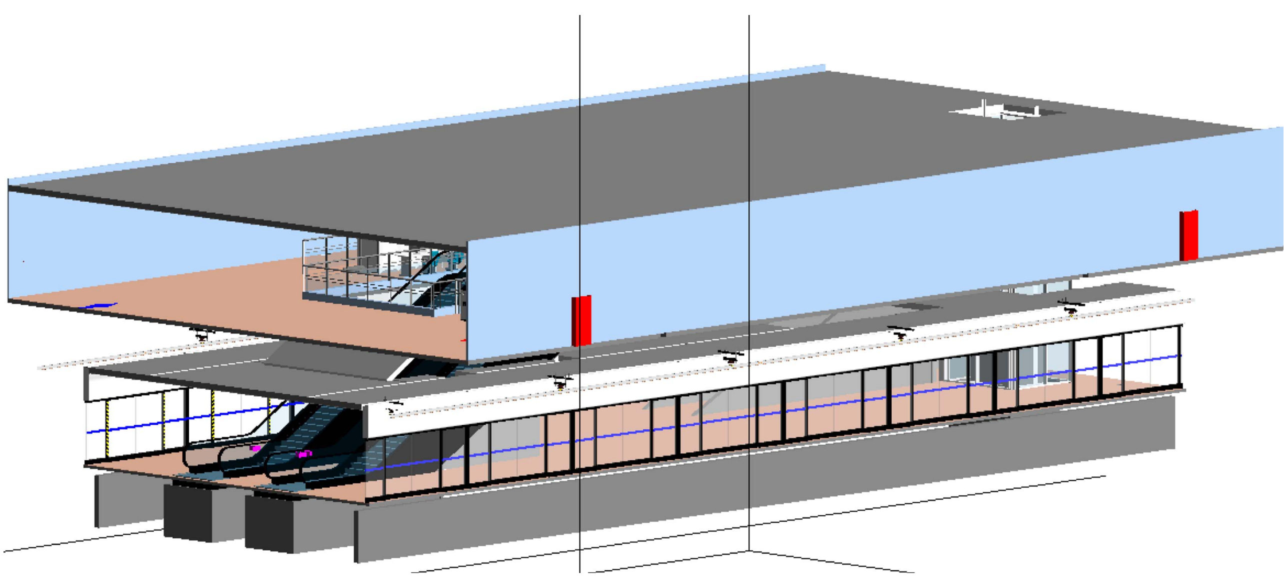
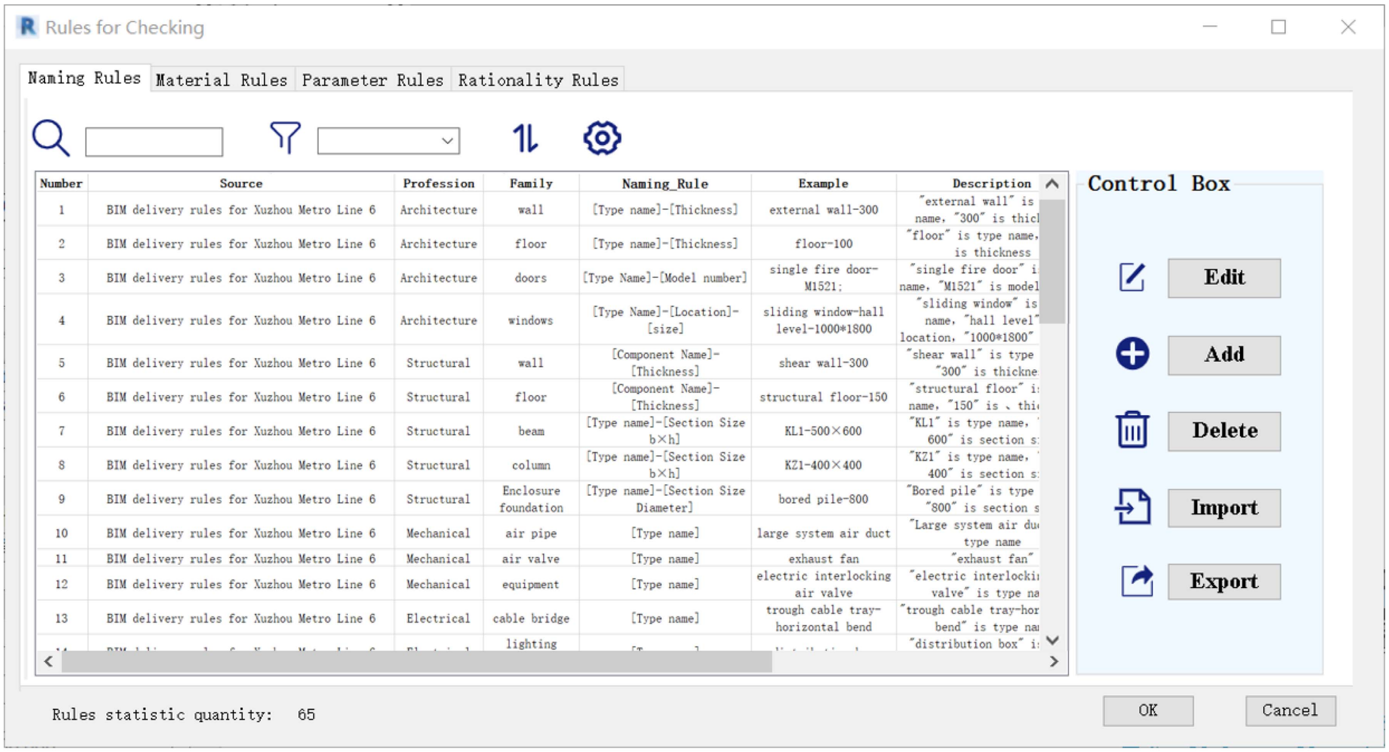


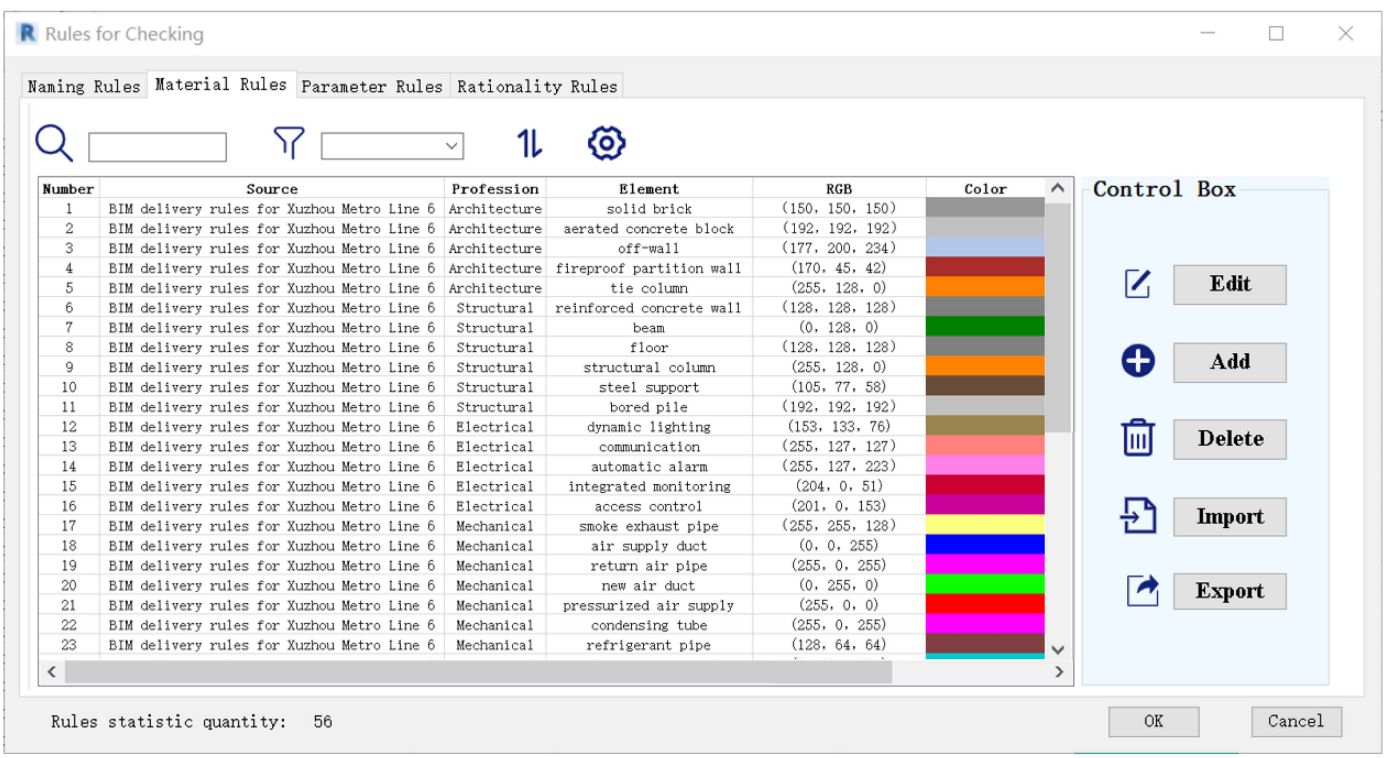
Fig. 10. BIM model of Xuzhou Metro Line 6 Beijing–Shanghai High-Speed Railway West Station.

improve the efficiency of personnel checks and the quality of BIM models.

Once the user clicks on a noncompliance message for a component, the model view will locate and highlight the specified component, and the user can modify the component attributes in response to the detailed description of the noncompliance presented in the report. To verify the accuracy of the checking system, the checking report of the noncompliant component is compared with

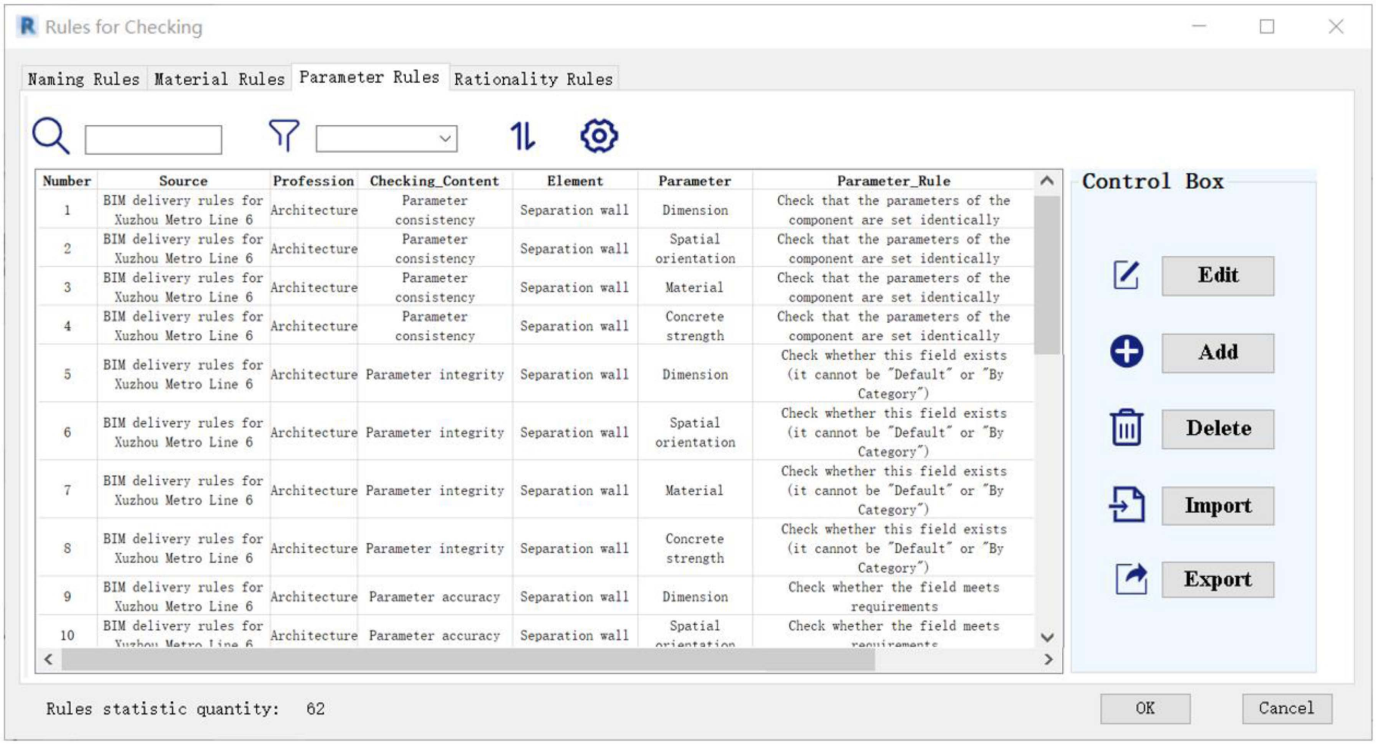


(a)

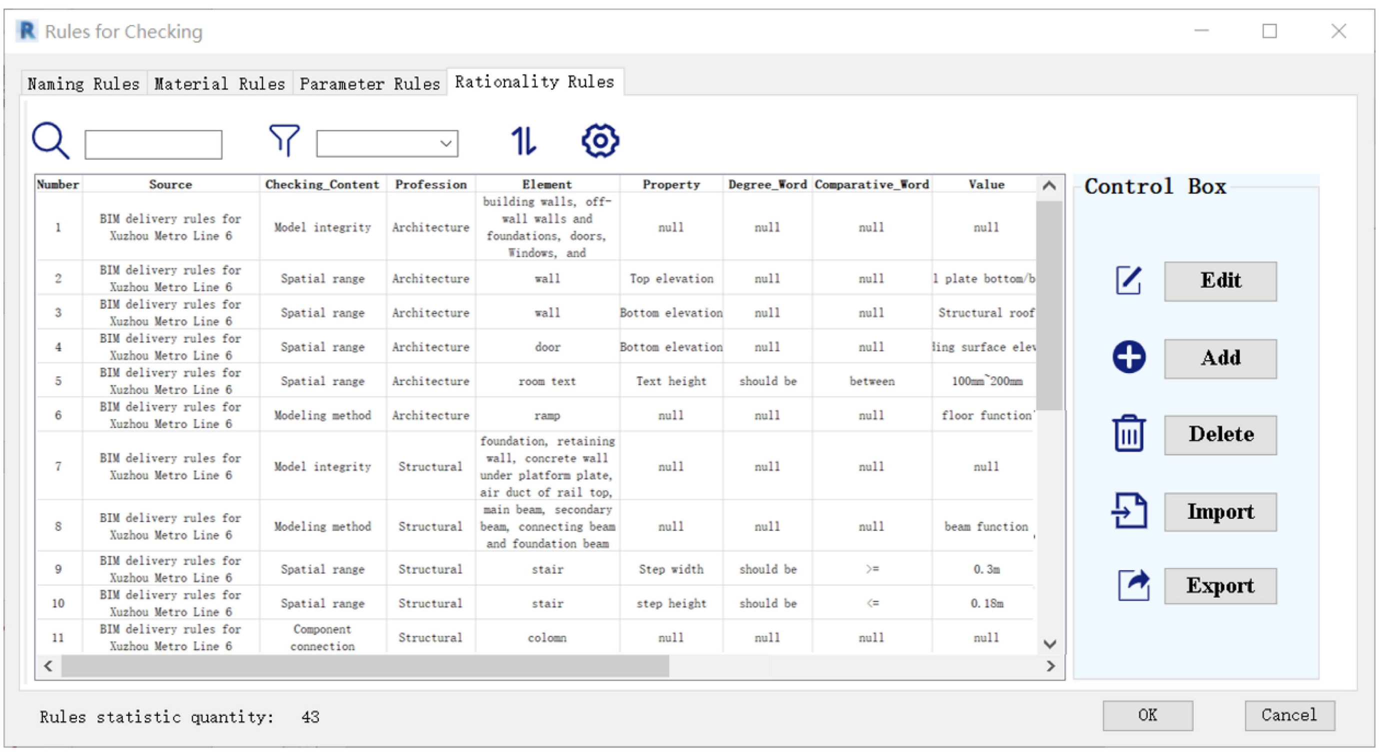


(b)

Fig. 11. Xuzhou Metro Line 6 rules knowledge base: (a) knowledge base of naming rules; (b) knowledge base of material rules; (c) knowledge base of parameter rules; and (d) knowledge base of rationality rules.



(c)



(d)

Fig. 11. (*Continued.*)

the component properties, and the results show that the checking results of the system are correct. For example, by clicking on the noncompliant escalator (ID 2035785), the system automatically locates and highlights the component, and the report shows that the naming of the component with the same name does not comply with the rules (Fig. 13). Specifically, the problem detail is "Naming does not comply with the rules: [Type Name]." The member's properties panel in Revit should be compared in order to make sure the checking result is correct.

Clicking on the floor member that does not comply with the rules (ID 1089624), the report shows that the member naming, material, and parameters do not comply with the rules (Fig. 14).

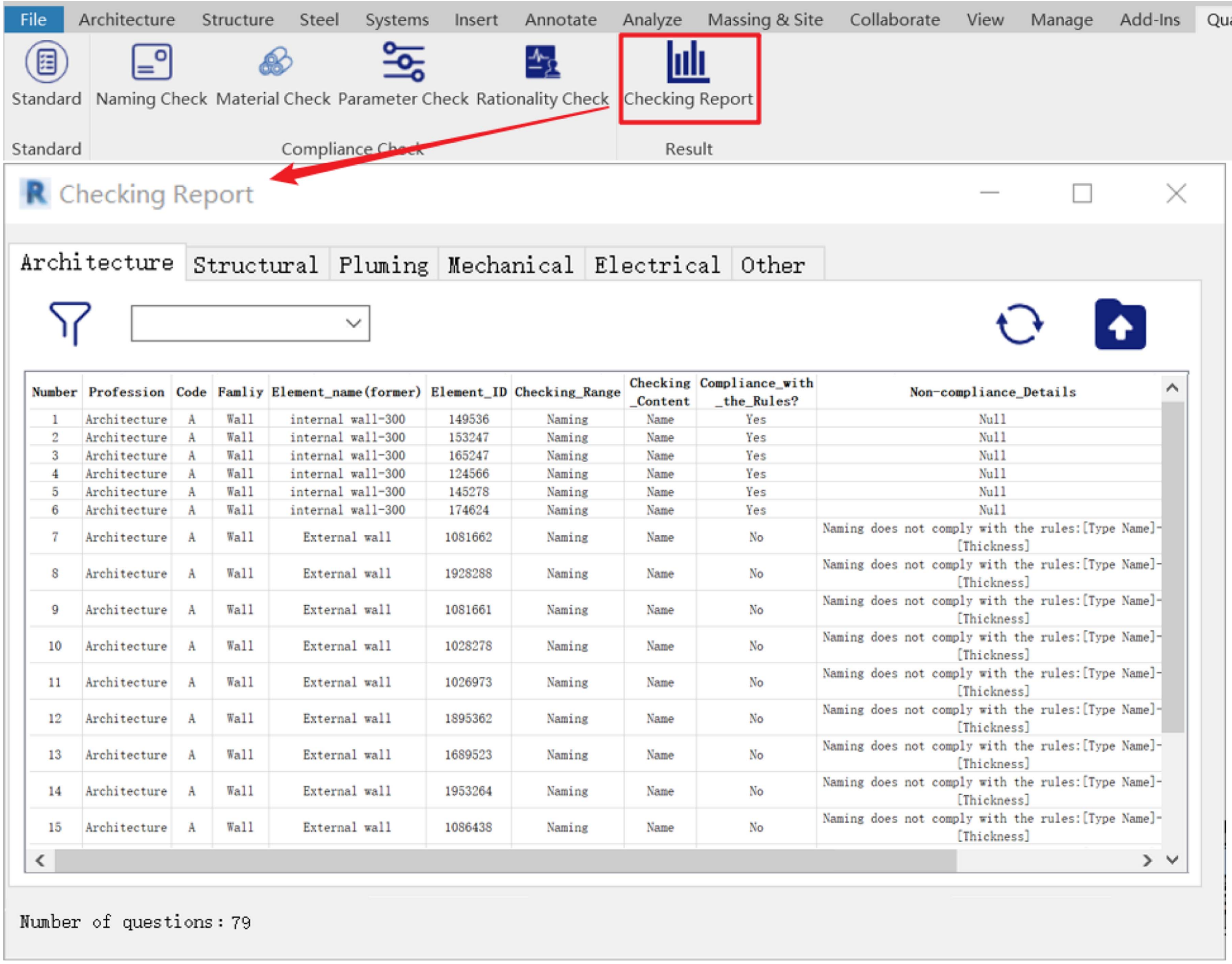


Fig. 12. Checking report for the architecture profession.

The noncompliance details are "The named thickness is inconsistent with the floor thickness parameter" and "The material does not comply with the rules: RGB (128, 128, 128)." Checking the properties panel of this component, we can see that its named thickness is "100" but the actual thickness parameter is "120," which is inconsistent. Moreover, the real color of the component is "RGB (0, 0, 0)," which really does not meet the rule requirements, and the checking system determines that it is correct. Therefore, comparing the consistency of the checking results with the Revit property can verify that the checking results are reliable.

Evaluation of System Performance Based on Checking Reports

The BIM model quality compliance checking system based on knowledge graph has been applied and tested in the BIM model case of Xuzhou Metro Line 6 Beijing–Shanghai High-Speed Railway West Station. The performance evaluation of the system mainly relies on an automatically generated checking report, which details various noncompliance issues in the model, including specific nonconformities in naming, materials, parameters, and rationality. According to the preceding analysis and statistics of the application results of the case, the superiority and feasibility of the performance of the functional components of the developed system are verified, mainly in the following aspects:

- Rule base coverage and flexibility: In the checking process of this case, the existing national BIM modeling standards and BIM delivery standards of this project are integrated, and a total of 112 named checking rules, material checking rules, parameter checking rules, and rationality checking rules are generated through the systematic identification and user addition of the rule knowledge base. It was used for the compliance checking of the BIM model of Xuzhou Metro Line 6 Beijing–Shanghai High-Speed Railway West Station.
- Performance efficiency: Compared with the traditional manual inspection, the system significantly improves checking efficiency and can complete the compliance checking of a large number of components in a short time.
- System identification ability and accuracy: The BIM model in this case has a total of 2,496 components, including 398 noncompliant components identified by the system, and multiple professional fields such as architecture, structural, mechanical, electrical, and plumbing. The system can accurately identify noncompliant components in the BIM model, such as the naming problems of escalators and the inconsistency between floor naming and real attribute parameters. The high accuracy and

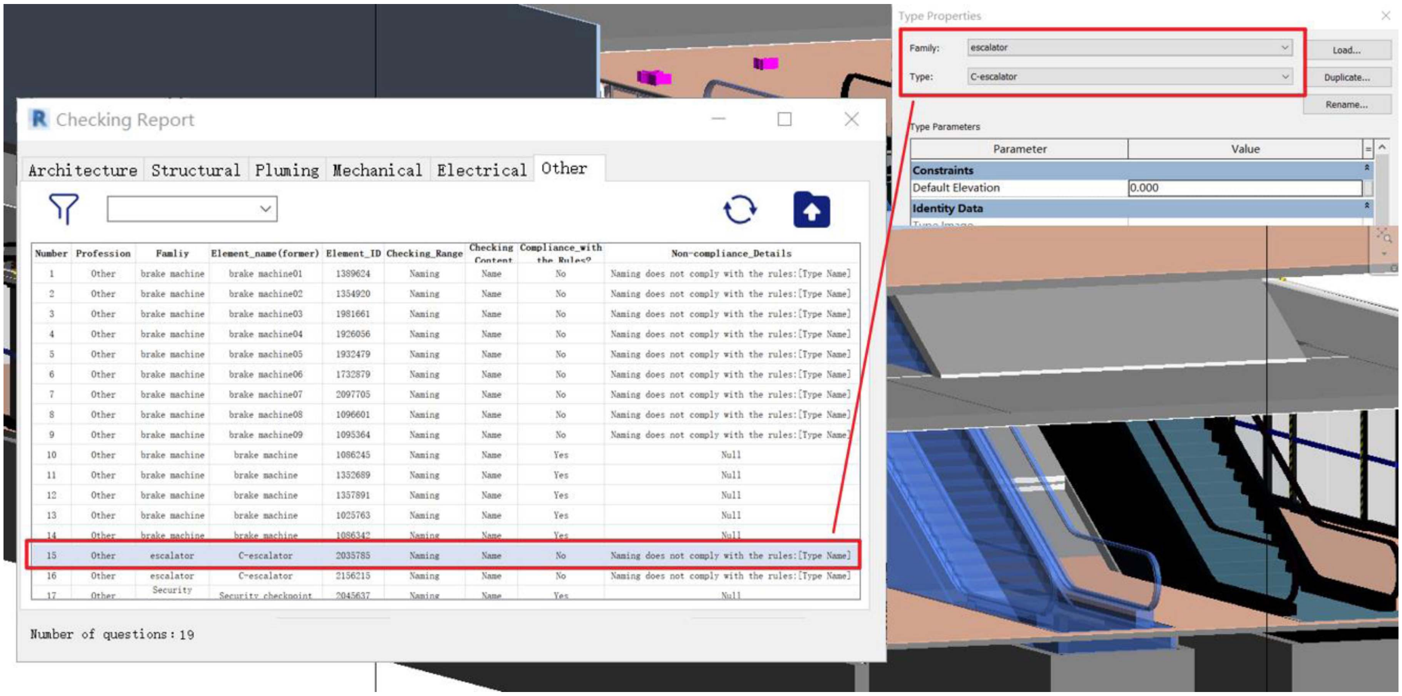


Fig. 13. Noncompliant escalator details.

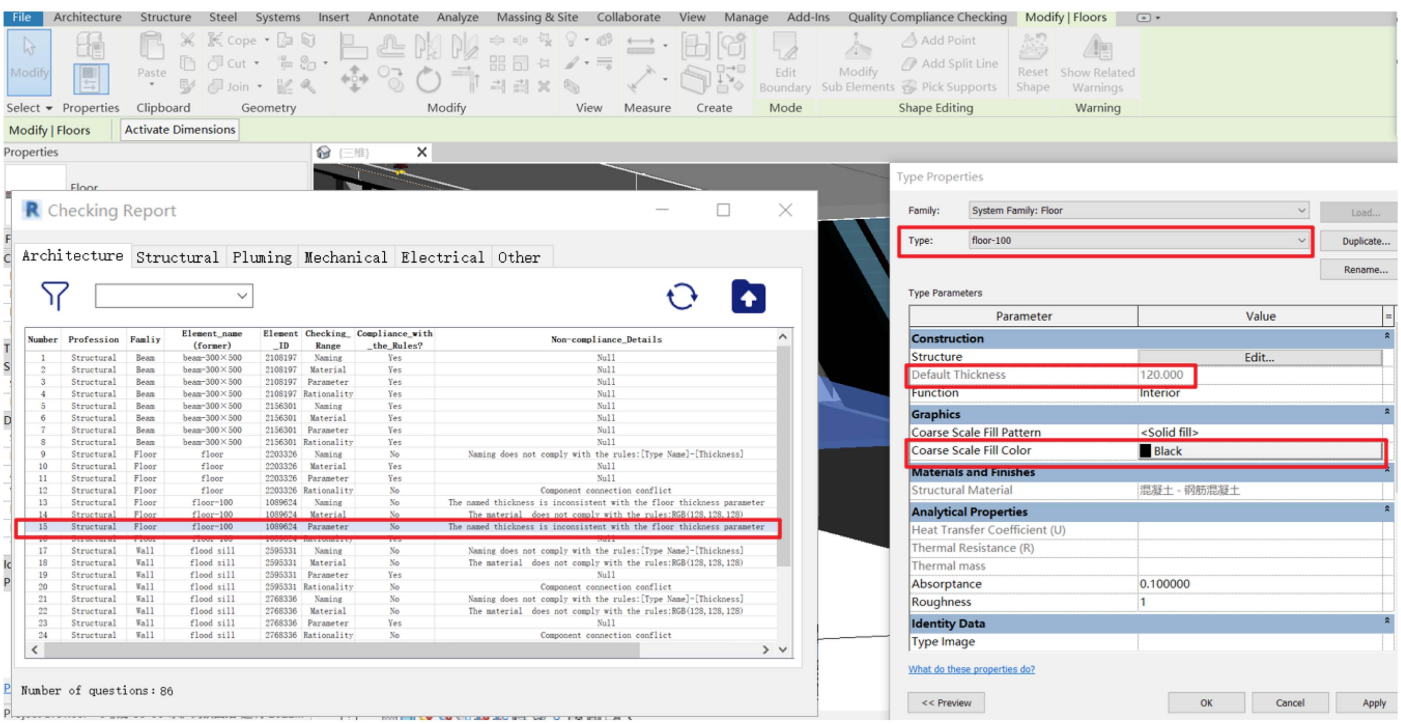


Fig. 14. Noncompliant floor details.

reliability of the system are verified by comparing the compliance check results with the consistency of Revit properties.

- User experience and operability: In Revit, the component ID has a unique identification role, so the system checking adopts the method of error component ID location, accurate to the ID of each component under check, the corresponding error type, and detailed checking results. The visual checking report and real-time update function provided by the system enable users to quickly locate problem components, significantly improving the work efficiency of inspectors and the quality of BIM models.
- Model correctness: The checkers modify the model component parameters and attributes by checking the results, and the number of model problems is updated. Based on the checking report, the model was gradually modified and improved, and the error rate of the model was reduced from the original 15.95% to 0%. The BIM model was completely correct and complied with all the rules.

Conclusion

BIM technology has brought strong technical support to the information management of construction projects and promoted the deep integration of engineering design and information technology. Normative modeling is the basis to ensure the quality of information models. Due to the low standardization degree of existing BIM models and the low efficiency and poor accuracy of manual checking methods, the main purposes of this study are to improve the modeling correctness of BIM models, develop an automated BIM model quality checking system, and enhance the standardization level of BIM models. Focusing on this purpose, this study proposed a method of BIM model quality conformance checking based on knowledge graph technology, which has certain innovation and practicability.

This study has carried out the following:

- The research framework for the BIM model quality compliance checking system based on knowledge graph was established.
- Based on BIM standards, the text was structured and classified, and rules were established to construct knowledge graph. The dynamic rule knowledge base was developed based on C# and SQL, which breaks the limitation of hard coding and realizes the dynamic management of BIM standards.
- A BIM model quality checking system for the Revit platform was developed using C# language to realize automatic checking and quality assessment of BIM model components.
- Through the case application of a Xuzhou subway BIM model, the feasibility and practicability of the developed system were verified. The results showed that the system can automatically check and generate reports, accurately find the problems existing in the naming, materials, parameters, and rationality of components in a short time, and provide targeted improvement measures.

The main contributions of this paper are as follows: firstly, knowledge graph technology provides a structured method to represent and store the knowledge in the BIM modeling standard field, which improves the searchability and understandability of knowledge and facilitates the sharing and reuse of knowledge. Secondly, compared with the traditional manual checking method, the compliance checking of BIM models can be carried out by converting specifications into computer-recognizable language through knowledge graph technology, which can reduce the subjectivity of checks, improve the efficiency, accuracy, and digitization level of compliance checking, and enable modelers to create BIM models efficiently and correctly. Thirdly, the developed system can customize the rule base according to the BIM standards of different enterprises, apply it to different BIM projects to check the model, and propose modification suggestions to realize standardized modeling and improve the work efficiency of BIM designers.

However, the study had some limitations. First, as time goes by, BIM modeling standards and regulations are constantly updated, and the complexity of texts will affect whether they can be converted into formable rules. Therefore, higher technical requirements are put forward for the maintenance and update of the knowledge rule base. Secondly, because the BIM software used by various enterprises is not uniform and is affected by existing comprehensive factors (cost, time, and so on), the plug-in developed is only used for Revit software, and the checking function is limited to the check of the name, material, parameters, and others, and the reasoning of complex spatial relationships requires more in-depth research.

Therefore, future research work will be devoted to the following aspects: (1) studying how to realize the check of BIM models in different formats exported by different software, improve the checking function of BIM models, and improve the stability of algorithms; and (2) investigating how to automatically modify the properties or parameters of noncompliant components based on rules and checking reports. Future research will strive to create a more intelligent BIM model quality compliance checking system to provide support and guarantee for the successful implementation and management of construction projects.

Data Availability Statement

Some or all data, models, or code generated or used during the study are proprietary or confidential in nature and may only be provided with restrictions.

Acknowledgments

The authors sincerely acknowledge the editors and anonymous reviewers for their valuable comments and constructive suggestions, which considerably improved the exposition of this work. The authors also gratefully acknowledge those who provided data and suggestions. This work was funded by the Graduate Innovation Program of China University of Mining and Technology (Project No. 2023WLJCRCZL064), the Postgraduate Research & Practice Innovation Program of Jiangsu Province (Project No. KYCX23_2878), the National Natural Science Foundation of China (Grant No. 72171224), and the Humanities and Social Sciences Foundation of China's Education Ministry (Grant No. 19YJAZH122).

References

Bao, Q., J. Zhou, Y. Zhao, X. Li, S. Tao, and P. Duan. 2022. "Developing a rule-based dynamic safety checking method for enhancing construction safety." *Sustainability* 14 (21): 14130. https://doi.org/10.3390/su142114130.

Beach, T. H., Y. Rezgui, H. Li, and T. Kasim. 2015. "A rule-based semantic approach for automated regulatory compliance in the construction sector." *Expert Syst. Appl.* 42 (12): 5219–5231. https://doi.org/10.1016/j.eswa.2015.02.029.

Bloch, T., and R. Sacks. 2020. "Clustering information types for semantic enrichment of building information models to support automated code compliance checking." *J. Comput. Civ. Eng.* 34 (6): 04020040. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000922.

China Architectural Design Institute. 2016. *Chinese Academy of sciences BIM standard series—Naming rules*. Beijing: China Architectural Design Institute.

Choi, C. H., and J. Lee. 2022. "A BIM-based quality inspection system prototype for temporary construction works." *Buildings* 12 (11): 1931. https://doi.org/10.3390/buildings12111931.

Choi, J., J. Choi, and I. Kim. 2014. "Development of BIM-based evacuation regulation checking system for high-rise and complex buildings." *Autom. Constr.* 46 (Oct): 38–49. https://doi.org/10.1016/j.autcon.2013.12.005.

Chowdhary, K., and K. Chowdhary. 2020. "Natural language processing." In *Fundamentals of artificial intelligence*, 603–649. Draper, UT: Pluralsight.

Deng, H., Y. Xu, Y. Deng, and J. Lin. 2022. "Transforming knowledge management in the construction industry through information and communications technology: A 15-year review." *Autom. Constr.* 142 (Oct): 104530. https://doi.org/10.1016/j.autcon.2022.104530.

Eastman, C., J.-M. Lee, Y.-S. Jeong, and J.-K. Lee. 2009. "Automatic rule-based checking of building designs." *Autom. Constr.* 18 (8): 1011–1033. https://doi.org/10.1016/j.autcon.2009.07.002.

Esteves, J., and R. C. Joseph. 2008. "A comprehensive framework for the assessment of eGovernment projects." *Gov. Inf. Q.* 25 (1): 118–132. https://doi.org/10.1016/j.giq.2007.04.009.

Fan, S.-L., H.-L. Chi, and P.-Q. Pan. 2019. "Rule checking Interface development between building information model and end user." *Autom. Constr.* 105 (Sep): 102842. https://doi.org/10.1016/j.autcon.2019.102842.

Fenves, S. J. 1966. "Tabular decision logic for structural design." *J. Struct. Div.* 92 (6): 473–490. https://doi.org/10.1061/JSDEAG.0001567.

Guo, D., E. Onstein, and A. D. La Rosa. 2021. "A semantic approach for automated rule compliance checking in construction industry." *IEEE Access* 9 (Aug): 129648–129660. https://doi.org/10.1109/ACCESS.2021.3108226.

Ismail, A. S., K. N. Ali, and N. A. Iahad. 2017. "A review on BIM-based automated code compliance checking system." In *Proc., 2017 Int. Conf. on Research and Innovation in Information Systems (ICRIIS)*. London: IntechOpen Limited.

Jouve, D., Y. Amghar, B. Chabbat, and J.-M. Pinon. 2003. "Conceptual framework for document semantic modelling: An application to document and knowledge management in the legal domain." *Data Knowl. Eng.* 46 (3): 345–375. https://doi.org/10.1016/S0169-023X(03)00046-6.

Koo, B., and B. Shin. 2018. "Applying novelty detection to identify model element to IFC class misclassifications on architectural and infrastructure building information models." *J. Comput. Des. Eng.* 5 (4): 391–400. https://doi.org/10.1016/j.jcde.2018.03.002.

Li, X., D. Yang, J. Yuan, A. Donkers, and X. Liu. 2022. "BIM-enabled semantic web for automated safety checks in subway construction." *Autom. Constr.* 141 (Sep): 104454. https://doi.org/10.1016/j.autcon.2022.104454.

Liebich, T., J. Wix, and Z. Qi. 2004. *Speeding-up the submission process, The Singapore e-plan checking project offers automatic plan checking based on IFC*, 245–252. Reston, VA: ASCE.

Liu, B., X. Wang, P. Liu, S. Li, Q. Fu, and Y. Chai. 2021. "Unikg: A unified interoperable knowledge graph database system." In *Proc., IEEE 37th Int. Conf. on Data Engineering (ICDE)*. New York: IEEE.

Luo, H., and P. Gong. 2015. "A BIM-based code compliance checking process of deep foundation construction plans." *J. Intell. Rob. Syst.* 79 (Aug): 549–576. https://doi.org/10.1007/s10846-014-0120-z.

Martins, J. P., and A. Monteiro. 2013. "LicA: A BIM based automated code-checking application for water distribution systems." *Autom. Constr.* 29 (Jan): 12–23. https://doi.org/10.1016/j.autcon.2012.08.008.

Ministry of Housing and Urban-Rural Development of the People's Republic of China. 2016. *Unified standard for the application of building information models*. GB/T51212-2016. Beijing: Ministry of Housing and Urban-Rural Development of the People's Republic of China.

Ministry of Housing and Urban-Rural Development of the People's Republic of China. 2017a. *Building information model construction application standard*. GB/T51235-2017. Beijing: Ministry of Housing and Urban-Rural Development of the People's Republic of China.

Ministry of Housing and Urban-Rural Development of the People's Republic of China. 2017b. *Standard for classification and coding of building information model*. GB/T51269-2017. Beijing: Ministry of Housing and Urban-Rural Development of the People's Republic of China.

Ministry of Housing and Urban-Rural Development of the People's Republic of China. 2018. *Standard for the design and delivery of building information model*. GB/T51301-2018. Beijing: Ministry of Housing and Urban-Rural Development of the People's Republic of China.

Ministry of Housing and Urban-Rural Development of the People's Republic of China. 2021. *Building information model storage standard*. GB/T51447-2021. Beijing: Ministry of Housing and Urban-Rural Development of the People's Republic of China.

Nawari, N. O. 2019. "A generalized adaptive framework (GAF) for automating code compliance checking." *Buildings* 9 (4): 86. https://doi.org/10.3390/buildings9040086.

Pauwels, P., R. De Meyer, and J. Van Campenhout. 2010. "Interoperability for the design and construction industry through semantic web technology." In *Proc., Int. Conf. on Semantic and Digital Media Technologies*. Berlin: Springer.

Peng, J., and X. Liu. 2023. "Automated code compliance checking research based on BIM and knowledge graph." *Sci. Rep.* 13 (1): 7065. https://doi.org/10.1038/s41598-023-34342-1.

Salama, D. M., and N. M. El-Gohary. 2016. "Semantic text classification for supporting automated compliance checking in construction." *J. Comput. Civ. Eng.* 30 (1): 04014106. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000301.

Shen, Q., S. Wu, Y. Deng, H. Deng, and J. C. Cheng. 2022. "BIM-based dynamic construction safety rule checking using ontology and natural language processing." *Buildings* 12 (5): 564. https://doi.org/10.3390/buildings12050564.

Solihin, W., J. Dimyadi, Y.-C. Lee, C. Eastman, and R. Amor. 2017. "The critical role of accessible data for BIM-based automated rule checking systems." In *Proc., Joint Conf. on Computing in Construction (JC3)*. Edinburgh, UK: Heriot-Watt Univ. https://doi.org/10.24928/JC3-2017/0161.

Solihin, W., J. Dimyadi, Y.-C. Lee, C. Eastman, and R. Amor. 2020. "Simplified schema queries for supporting BIM-based rule-checking applications." *Autom. Constr.* 117 (Sep): 103248. https://doi.org/10.1016/j.autcon.2020.103248.

Solihin, W., and C. M. Eastman. 2016. "A knowledge representation approach in BIM rule requirement analysis using the conceptual graph." *J. Inf. Technol. Constr.* 21 (Jun): 370–401.

Sun, H., and I. Kim. 2022. "Automated checking system for modular BIM objects." *J. Civ. Eng. Manage.* 28 (7): 554–563. https://doi.org/10.3846/jcem.2022.17230.

Wang, X., and N. El-Gohary. 2022. "Deep learning-based relation extraction from construction safety regulations for automated field compliance checking." In *Proc., Construction Research Congress 2022*. Reston, VA: ASCE.

Wang, Y., L. Zhang, H. Yu, and R. L. Tiong. 2022. "Detecting logical relationships in mechanical, electrical, and plumbing (MEP) systems with BIM using graph matching." *Adv. Eng. Inf.* 54 (Oct): 101770. https://doi.org/10.1016/j.aei.2022.101770.

Wangara, J. 2018. "Quality management in BIM: Use of Solibri model checker and CoBIM guidelines for BIM quality validation." Bachelor's thesis, Dept. of Engineering, Helsinki Metropolia Univ. of Applied Sciences.

Xie, X., J. Zhou, X. Fu, R. Zhang, H. Zhu, and Q. Bao. 2022. "Automated rule checking for MEP systems based on BIM and KBMS." *Buildings* 12 (7): 934. https://doi.org/10.3390/buildings12070934.

Yan, J., C. Wang, W. Cheng, M. Gao, and A. Zhou. 2018. "A retrospective of knowledge graphs." *Front. Comput. Sci.* 12 (Feb): 55–74. https://doi.org/10.1007/s11704-016-5228-9.

Yang, M., Q. Zhao, L. Zhu, H. Meng, K. Chen, Z. Li, and X. Hei. 2023. "Semi-automatic representation of design code based on knowledge graph for automated compliance checking." *Comput. Ind.* 150 (Sep): 103945. https://doi.org/10.1016/j.compind.2023.103945.

Zhang, J., and N. M. El-Gohary. 2016. "Semantic NLP-based information extraction from construction regulatory documents for automated compliance checking." *J. Comput. Civ. Eng.* 30 (2): 04015014. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000346.

Zhang, J., and N. M. El-Gohary. 2017. "Integrating semantic NLP and logic reasoning into a unified system for fully-automated code checking." *Autom. Constr.* 73 (Jan): 45–57. https://doi.org/10.1016/j.autcon.2016.08.027.

Zhang, R., and N. El-Gohary. 2021. "A deep neural network-based method for deep information extraction using transfer learning strategies to support automated compliance checking." *Autom. Constr.* 132 (Dec): 103834. https://doi.org/10.1016/j.autcon.2021.103834.

Zhang, S., F. Boukamp, and J. Teizer. 2015. "Ontology-based semantic modeling of construction safety knowledge: Towards automated safety planning for job hazard analysis (JHA)." *Autom. Constr.* 52 (Apr): 29–41. https://doi.org/10.1016/j.autcon.2015.02.005.

Zhou, P., and N. El-Gohary. 2017. "Ontology-based automated information extraction from building energy conservation codes." *Autom. Constr.* 74 (Feb): 103–117. https://doi.org/10.1016/j.autcon.2016.09.004.